Homework #2 for Geospatial Analysis & Remote Sensing (EDS223 - Fall 2026)

2026-10-15

```
# Load in Packages
library(tidyverse)
library(here)
library(sf)
library(tmap)


# Load in data
ej <- sf::st_read(here::here(
  "data",
  "ejscreen",
  "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"
))

LA_crashes <- sf::st_read(here::here(
  "data",
  "SWITRS_LA",
  "Crashes.csv"
))

map_inequal <- sf::st_read(here::here(
  "data",
  "mapping-inequality",
  "mapping-inequality-los-angeles.json"
))


# Filter EjScreen data
California_ej <- ej |>
```

```
  dplyr::filter(ST_ABBREV == "CA")


# Filter dataset to a county
LA_county_ej <- California_ej |>
  dplyr::filter(CNTY_NAME == "Los Angeles County")
```

```
# Check coordinate systems for compatibility
st_crs(LA_county_ej)
st_crs(LA_crashes)
st_crs(map_inequal)


# Quick test to see
st_crs(LA_county_ej) == st_crs(map_inequal)


# Change coorindate systems
map_inequal_transform <- st_transform(map_inequal, crs = st_crs(LA_county_ej))


# Check to see CRS's match
if (st_crs(map_inequal_transform) == st_crs(LA_county_ej)) {
  print("it's a match!")
} else {
  print("still not a match")
}
```

Part 1. Legacy of redlining in current environmental (in)justice

1. Create a map of historical redlining neighborhoods

```
# Map of Legacy Redlining in LA County

# Specifying dataframe from filtered data
tm_shape(LA_county_ej) +

  # Specify shape and customize specific attributes to enhance aesthetic
  tm_polygons(
    lwd = 0.15,
```

```
  col = "black"
) +

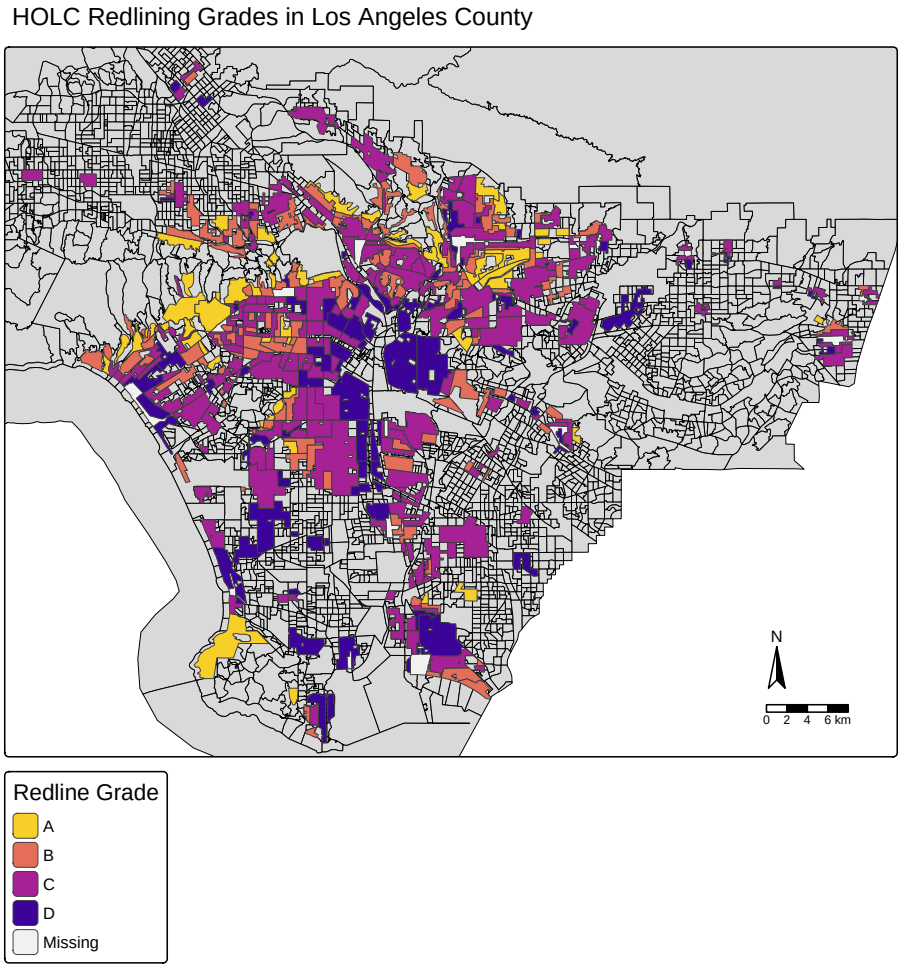
# Specifying dataframe from filtered data and
# set layer as main for map extent
tm_shape(map_inequal_transform, is.main = TRUE) +

# Specify shape and customize specific attributes to enhance aesthetic
tm_polygons(
  fill = "grade",
  lwd = 0.2,
  fill.legend = tm_legend(title = "Redline Grade"),
  fill.scale = tm_scale(values = "-plasma")
) +

# Add a north arrow in the bottom right corner
tm_compass(size = 2, text.size = 0.7, position = c("right", "bottom")) +

# Adding a scalebar in the bottom right corner
tm_scalebar(width = 8, text.size = 0.5, position = c("right", "bottom")) +

# Add a title above the map, positioned to the left
tm_title(
  "HOLC Redlining Grades in Los Angeles County",
  size = 1.0,
  position = tm_pos_out("center", "top", pos.h = "left", pos.v = "bottom")
)
```



2. Create a table

```
# Perform left join on the datasets
df_joined <- st_join(LA_county_ej, map_inequal_transform) |>
  st_drop_geometry()

# Count the block groups
block_percent <- df_joined |>
  count(grade) |>
  mutate(percent = round(n / sum(n) * 100, 2))
```

```
block_percent
```

```
  grade    n percent
1     A  449    5.00
2     B 1239   13.79
3     C 3058   34.02
4     D 1346   14.98
5  <NA> 2896   32.22
```